

FOSD — Unit 1

OS Fundamentals & Process Management

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- The question pages below are taken directly from the original source PDFs supplied in this conversation.
- Original question wording, options, tables, code, and figures are preserved on those pages.
- A complete original page is retained whenever a selected question depends on a figure/diagram, so the required visual context is not lost.
- The selected question numbers are listed below; neighboring questions may remain visible because the original page is preserved.
- The supplied 2017 Set-1 PDF is corrupted/unreadable in the uploaded ZIP, so no 2017 question is fabricated or substituted here.

Included source pages

CS2019 — source page 7 — selected Q7
CS2020 — source page 6 — selected Q2
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Q.17 The following C program is executed on a Unix/Linux system:

```
#include <unistd.h>
int main()
{
    int i;
    for (i=0; i<10; i++)
        if (i%2 == 0) fork();
    return 0;
}
```

The total number of child processes created is _____.

Q.18 Consider the following C program:

```
#include <stdio.h>
int jumble(int x, int y){
    x=2*x+y;
    return x;
}
int main(){
    int x=2, y=5;
    y=jumble(y,x);
    x=jumble(y,x);
    printf("%d \n", x);
    return 0;
}
```

The value printed by the program is _____.

Q.19 Consider the grammar given below:

$S \rightarrow Aa$
 $A \rightarrow BD$
 $B \rightarrow b \mid \epsilon$
 $D \rightarrow d \mid \epsilon$

Let a, b, d, and \$ be indexed as follows:

a	b	d	\$
3	2	1	0

Compute the FOLLOW set of the non-terminal B and write the index values for the symbols in the FOLLOW set in the descending order. (For example, if the FOLLOW set is {a, b, d, \$}, then the answer should be 3210)

Answer: _____.

Consider the language $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and the following statements.

- I. L is deterministic context-free.
- II. L is context-free but not deterministic context-free.
- III. L is not $LL(k)$ for any k .

Which of the above statements is/are TRUE?

- (A) I only
- (B) II only
- (C) I and III only
- (D) III only

Q.No. 11 Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?

- (A) The hole created by first fit is always larger than the hole created by next fit.
- (B) The hole created by worst fit is always larger than the hole created by first fit.
- (C) The hole created by best fit is never larger than the hole created by first fit.
- (D) The hole created by next fit is never larger than the hole created by best fit.

Q.No. 12 Consider the following statements about process state transitions for a system using preemptive scheduling.

- I. A running process can move to ready state.
- II. A ready process can move to running state.
- III. A blocked process can move to running state.
- IV. A blocked process can move to ready state.

Which of the above statements are TRUE?

- (A) I, II, and III only
- (B) II and III only
- (C) I, II, and IV only
- (D) I, II, III, and IV

Q.No. 13



Q.13	Suppose a database system crashes again while recovering from a previous crash. Assume checkpointing is not done by the database either during the transactions or during recovery. Which of the following statements is/are correct?
(A)	The same undo and redo list will be used while recovering again.
(B)	The system cannot recover any further.
(C)	All the transactions that are already undone and redone will not be recovered again.
(D)	The database will become inconsistent.

Q.14	Which of the following standard C library functions will <i>always</i> invoke a system call when executed from a single-threaded process in a UNIX/Linux operating system?
(A)	<code>exit</code>
(B)	<code>malloc</code>
(C)	<code>sleep</code>
(D)	<code>strlen</code>

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Q.19	Consider the following threads, T_1 , T_2 , and T_3 executing on a single processor, synchronized using three binary semaphore variables, S_1 , S_2 , and S_3 , operated upon using standard <code>wait()</code> and <code>signal()</code> . The threads can be context switched in any order and at any time.		
	T_1 <pre>while(true){ wait(S_3); print("C"); signal(S_2); }</pre>	T_2 <pre>while(true){ wait(S_1); print("B"); signal(S_3); }</pre>	T_3 <pre>while(true){ wait(S_2); print("A"); signal(S_1); }</pre>
	Which initialization of the semaphores would print the sequence BCABCABCA....?		
(A)	$S_1 = 1; S_2 = 1; S_3 = 1$		
(B)	$S_1 = 1; S_2 = 1; S_3 = 0$		
(C)	$S_1 = 1; S_2 = 0; S_3 = 0$		
(D)	$S_1 = 0; S_2 = 1; S_3 = 1$		

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Q.22	Which one or more of the following need to be saved on a context switch from one thread (T1) of a process to another thread (T2) of the same process?
(A)	Page table base register
(B)	Stack pointer
(C)	Program counter
(D)	General purpose registers

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Q.23	Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode?
(A)	Function Call
(B)	<code>malloc</code> Call
(C)	Page Fault
(D)	System Call

Q.23 Let L_1, L_2 be two regular languages and L_3 a language which is not regular. Which of the following statements is/are *always* TRUE?

- (A) $L_1 = L_2$ if and only if $L_1 \cap \overline{L_2} = \phi$
- (B) $L_1 \cup L_3$ is not regular
- (C) $\overline{L_3}$ is not regular
- (D) $\overline{L_1} \cup \overline{L_2}$ is regular

Q.24 Which of the following statements about threads is/are TRUE?

- (A) Threads can only be implemented in kernel space
- (B) Each thread has its own file descriptor table for open files
- (C) All the threads belonging to a process share a common stack
- (D) Threads belonging to a process are by default not protected from each other

Q.25 Which of the following process state transitions is/are NOT possible?

- (A) Running to Ready
- (B) Waiting to Running
- (C) Ready to Waiting
- (D) Running to Terminated

Q.26 Which of the following is/are Bottom-Up Parser(s)?

- (A) Shift-reduce Parser
- (B) Predictive Parser
- (C) LL(1) Parser
- (D) LR Parser

- Q.57 Consider the following code snippet using the `fork()` and `wait()` system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;
while(x > 0) {
    fork();
    printf("hello");
    wait(NULL);
    x--;
}
```

The total number of times the `printf` statement is executed is _____

- Q.58 Consider the entries shown below in the forwarding table of an IP router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation “/N” in a prefix indicates a subnet mask with the most significant N bits set to 1.

Prefix	Next hop router
10.1.1.0/24	R1
10.1.1.128/25	R2
10.1.1.64/26	R3
10.1.1.192/26	R4

This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are 10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191, and 10.1.1.205. The number of packets forwarded via the next hop router R2 is _____

- Q.59 Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S . The string $w = a^{30}b^{30}c^{30}$ is derivable from S . The number of steps (application of rules) in the derivation $S \rightarrow^* w$ is _____

Q.29	Suppose in a multiprogramming environment, the following C program segment is executed. A process goes into I/O queue whenever an I/O related operation is performed. Assume that there will always be a context switch whenever a process requests for an I/O, and also whenever the process returns from an I/O. The number of times the process will enter the ready queue during its lifetime (not counting the time the process enters the ready queue when it is run initially) is _____. (Answer in integer)
	<pre>int main() { int x=0,i=0; scanf("%d",&x); for(i=0; i<20; i++) { x = x+20; printf("%d\n",x); } return 0; }</pre>

Q.63	Consider the following program snippet. Assume that the program compiles and runs successfully. Further, assume that the <code>fork()</code> system call is always successful in creating a process. <pre> int main () { int i; for (i = 0; i < 3; i++){ if (fork() == 0){ continue; } break; } printf("Hello!"); return 0; } </pre> The total number of times that the <code>printf</code> statement gets executed is _____. (answer in integer)
Q.64	Consider a CPU that has to execute two types of processes. The first type, Actuators (A), requires a CPU burst of 6 seconds. The second type, Controllers (C), requires a CPU burst of 8 seconds. A new process of type A arrives at time $t = 10, 20, 30, 40,$ and 50 (in seconds). Similarly, a new process of type C arrives at time $t = 11, 22, 33, 44,$ and 55 (in seconds). The CPU scheduling policy is First Come First Serve (FCFS). The first process of type A starts running at $t = 10$ seconds. The average waiting time (in seconds) for the 10 processes is _____. (rounded off to one decimal place)
Q.65	Consider a relational database schema with a relation $R(A, B, C, D)$. If $\{A, B\}$ and $\{A, C\}$ are the only two candidate keys of the relation R, then the number of superkeys of relation R is _____. (answer in integer)